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21ECC402L - Project Review - 1

On

Intelligent DDoS Detection and Automated Mitigation for IoT–Cloud Networks Using Dynamic Fuzzing and Graph Neural Networks

by

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Flow of presentation

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Introduction

NetShield-GNN is a research framework for **Distributed Denial-of-Service (DDoS)** detection and mitigation in IoT–edge–cloud networks, where large fleets of low-power sensors, cameras and controllers are easily compromised and recruited into botnets. Because such attacks are distributed and individually low-rate, the evidence rarely sits in any single device; it lies in the relationships between devices, access routers and the edge gateway. The system therefore simulates the network packet by packet, exports **IPFIX-style flow records** every five seconds, converts each window into a graph, and classifies every node as Benign, Suspicious or Malicious with a **GraphSAGE graph neural network**. Detections drive a graduated rate-limit, block or isolate policy at the **edge gateway** inside the running simulation, so recovery is measured rather than assumed, enabling safe and reproducible evaluation before real deployment.

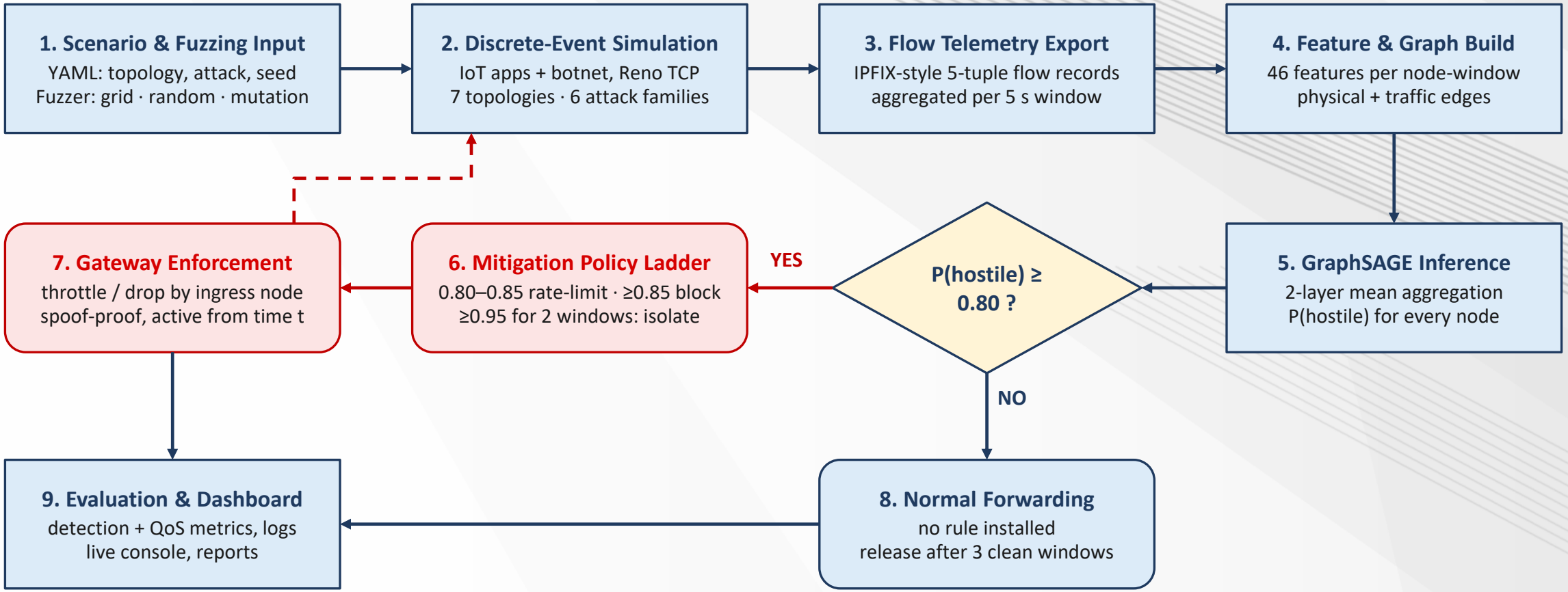
Problem Statement

IoT botnets launch DDoS attacks that saturate the edge-to-cloud uplink and exhaust server resources. Conventional **per-device threshold detectors** examine each device in isolation, so distributed low-rate, pulsing and slowloris attacks evade them, while legitimate surges such as firmware updates raise false alarms. Testing automated mitigation on a live network is also unsafe. **A detector is needed that exploits the network's relational structure and is validated in a closed loop without risk to real infrastructure.**

Objectives

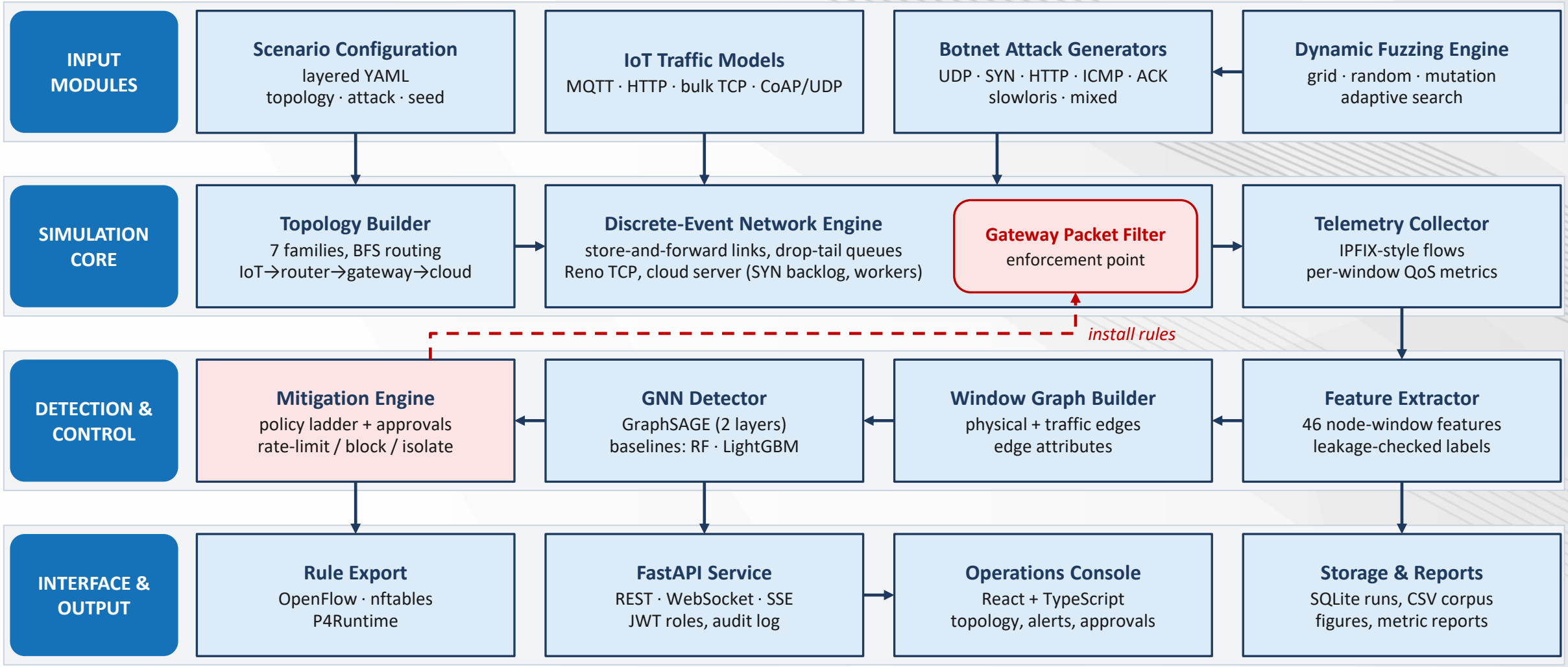
- **Packet-Level Network Simulation:** Build an in-memory discrete-event IoT–edge–cloud simulator covering seven topology families and six DDoS attack families.
- **Graph-Based Detection:** Model each 5 s window as a graph and classify every node as Benign, Suspicious or Malicious using GraphSAGE.
- **Closed-Loop Mitigation:** Rate-limit, block or isolate flagged devices at the gateway while the attack is still running, protecting clean-device QoS.
- **Robustness via Dynamic Fuzzing:** Generate unseen attack-parameter combinations (rate, size, pulsing, botnet size) to stress-test the detectors.
- **Comparative Evaluation:** Benchmark GraphSAGE against threshold, Random Forest and LightGBM detectors with run-grouped cross-validation.
- **Real-Time Monitoring:** Expose live telemetry, alerts and mitigation approvals through a REST/WebSocket API and an operations console.

Methodology



Dashed red arrow: rules installed at window t change packet handling inside the running simulation from t onwards (closed loop).

Block Diagram



Tools to be used

- **Python 3.10+:** core language for the simulator, feature pipeline, training and evaluation scripts.
- **PyTorch & PyTorch Geometric:** build and train the GraphSAGE, GATv2 and E-GraphSAGE models.
- **scikit-learn & LightGBM:** Random Forest and gradient-boosted baselines, and evaluation metrics.
- **NumPy & pandas:** flow aggregation, node-window feature matrices and run-grouped dataset splits.
- **NS-3 (via WSL2) & NetAnim:** cross-check scenarios in a standard simulator, with animation.
- **FastAPI & Uvicorn:** REST and WebSocket API serving live simulation state, alerts and approvals.
- **React + TypeScript (Vite):** operations console for topology, alerts, mitigation and replay.
- **Optuna:** adaptive (TPE) adversary that searches for evasive attack parameters.

Performance Metrics studied

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Detection Accuracy & Reliability

- **Macro-F1 Score:** mean per-class F1 over Benign / Suspicious / Malicious; robust to class imbalance.
- **Attack Recall (%):** share of hostile node-windows that are detected.
- **False-Positive Rate (%):** benign node-windows wrongly flagged; drives collateral blocking.
- **PR-AUC & ECE:** ranking quality and calibration of P(hostile), which the policy thresholds.

Network QoS & Mitigation

- **Time to Detect / Mitigate / Recover (s):** attack start to first flag, first rule and restored service.
- **Throughput (Mbit/s) & Uplink Utilisation (%):** goodput reaching the cloud and bottleneck load.
- **Latency p50/p99 (ms) & Jitter (ms):** one-way delay of clean devices (RFC 3550 jitter).
- **Loss Ratio & SLO Availability (%):** clean packets lost, or delivered within the 100 ms SLO.
- **Collateral Byte Ratio (%):** clean-device traffic dropped by mitigation.

Expected Results

- **Detection Within One Window:** attacking bots should be flagged at the first 5 s window boundary after onset, the design floor of a windowed detector.
- **Relational Detection of Stealthy Attacks:** GraphSAGE is expected to find distributed low-rate and slowloris bots that per-device thresholds miss.
- **Low False-Alarm Rate:** flash crowds, firmware and reboot storms should not trigger mitigation of legitimate devices.
- **Closed-Loop Recovery:** clean-device loss and latency should return towards the pre-attack baseline while the attack is still running.
- **Minimal Collateral Damage:** ingress-keyed rules should restrict only compromised nodes, even when source addresses are spoofed.
- **Structural Generalisation:** detection should hold on topology families withheld from training.

Future Work

- **Physical Testbed Validation:** run the agent on a Mininet / Open vSwitch testbed and on ESP32 or Raspberry Pi nodes, installing real OpenFlow rules.
- **In-Loop NS-3 Enforcement:** integrate the GNN detector into the NS-3 model so protocol-exact attacks are mitigated inside NS-3 runs.
- **Learned Mitigation Policy:** replace fixed probability thresholds with reinforcement learning driven by measured QoS feedback.
- **Federated & Streaming Deployment:** train jointly across gateways and serve inference through a Redis Streams + ONNX Runtime pipeline.
- **Real-Traffic Evaluation:** extend validation to public captures such as CICIoT2023, adding labels for each device and calibration for each site.

Conclusion

- **Problem:** IoT DDoS attacks are distributed, so per-device thresholds miss low-rate bots and misfire on legitimate surges.
- **Proposed Solution:** NetShield-GNN classifies every node of a per-window network graph with GraphSAGE, using packet-level simulation as its data source.
- **Closed-Loop Defence:** detections drive a graduated rate-limit, block and isolate policy at the gateway, measured inside the same run.
- **Expected Contribution:** a safe, reproducible platform and protocol to compare graph-based and conventional detectors on accuracy, timeliness and QoS.

References

[1] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, Long Beach, CA, USA, 2017, pp. 1024–1034.

[2] W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A graph neural network based intrusion detection system for IoT," in *Proc. IEEE/IFIP Netw. Oper. Manage. Symp. (NOMS)*, Budapest, Hungary, 2022, pp. 1–9.

[3] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and other botnets," *Computer*, vol. 50, no. 7, pp. 80–84, 2017.

[4] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, Art. no. 5941, 2023.

[5] G. F. Riley and T. R. Henderson, "The ns-3 network simulator," in *Modeling and Tools for Network Simulation*, K. Wehrle, M. Güneş, and J. Gross, Eds. Berlin, Germany: Springer, 2010, pp. 15–34.

[6] B. Claise, B. Trammell, and P. Aitken, "Specification of the IP Flow Information Export (IPFIX) protocol for the exchange of flow information," *IETF RFC 7011*, Sep. 2013.